

Ethan Reinhart

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Education

University of Florida

Bachelor of Science in Aerospace Engineering

GPA: 3.62

Relevant coursework: Computer-Aided Graphics and Design, Numerical Methods, Control of Dynamic Systems, Dynamics, Aerodynamics, Astrodynamics

Gainesville, Florida

Aug. 2023 - May 2027

Engineering Experience

Autonomous Rendezvous and Capture Research

May. 2026- Present

- Investigate autonomous rendezvous, tether-assisted capture, and post-capture guidance and control of non-cooperative spacecraft.
- Developed and simulated feedback control strategies for post-capture spacecraft despin and stabilization using tether endpoint actuation.
- Developed a lumped-mass spring-damper tether model incorporating orbital relative dynamics to simulate flexible-body behavior and spacecraft interactions.

Florida Rocket Lab - Guidance, Navigation and Control Team

Jan. 2025 -Present

- Design and evaluate PID and LQI control architectures for VTOL vehicle stabilization using MATLAB/Simulink.
- Developed Python flight-dynamics simulations for a liquid-propellant competition rocket and validated predicted trajectories against OpenRocket/RASAero II and flight data.
- Refined aerodynamic and trajectory models to predict competition flight performance within ~5% of flight results.
- Collaborate across GNC, propulsion, and vehicle teams to support modeling, simulation, testing, and system integration.

Projects

Spacecraft Rendezvous and Proximity Operations Simulation

Jan. 2025 - May 2025

- Developed a Python simulation of spacecraft rendezvous and relative motion using Clohessy-Wiltshire dynamics in the LVLH reference frame.
- Implemented LQI-based guidance and control to drive the chaser toward a target relative position and velocity.

Flight Control Design for VTOL Drone

- Designed and implemented PID and LQI flight-control algorithms for an autonomous VTOL vehicle using MATLAB/Simulink.
- Modeled nonlinear vehicle dynamics and implemented actuator command mapping and saturation constraints for embedded flight control.
- Validated controller performance through simulation and physical flight testing, achieving stable autonomous flight and landing.
- Co-authored and presented the control-system design and experimental results at the 2025 AIAA Student Conference.

Work Experience

Food Server

Ichiban Sushi Gainesville

Mar. 2022 - Dec. 2024

- Provided customer service and multitasked in a fast-paced, high pressure environment
- Collaborated with team members to ensure efficient operations.

Additional Skills

- **Programming:** Python, MATLAB
- **Engineering Software:** Simulink, SOLIDWORKS, OpenRocket, RASAero II
- **Tools:** Git, Microsoft Excel
- **Controls & GNC:** PID, LQR/LQI, State-Space Control, Feedback Control
- **Astrodynamics:** Clohessy-Wiltshire Dynamics, Orbital Mechanics, Relative Motion, Numerical Simulation